

PISA's PL4 Format Time domain

Leonardo M. N. de Mattos edited this page on Mar 5, 2020 · [2 revisions](#)

Binary file byte's positions for time domain simulations

	"columns"																
"Row"	8 bytes							8 bytes							Notes		
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
0																	ASCII HEADER
1																	Seems to be some kind of versioning
2	00	00	00	00	?TMAX?			DELTAT									
3	#VAR X2							PL4 SIZE+1			00	00	00	00			
4	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	"Row" of zeroes before var names
5	20	20	20	VT	Node From				Node To								
	...															Begins variables, one for each "row", <u>ascii</u> encoding.	
5+n	20	20	20	VT	Node From				Node To								
5+n+1	t[0]			var1[0]			var2[0]			var3[0]						Begins actual data, sequence of IEEE 754 float in <u>little endian</u> order	
	...																

- ?TMAX? - This field is related to TMAX, however it is not exactly TMAX. *Still needs further investigation*;
- DELTAT - 4 bytes IEEE 754 floating point, little endian;
- #VAR X2 - In this field there is twice the number of variables. Not including the time. (4 bytes integer);
- PL4 SIZE+1 - The PL4 file size in bytes + 1 byte (4 bytes integer);
- The 'row' 4 precedes the variables description. It is filled with zeroes;
 - Every variable 'row' starts with the sequence of 202020 in the first three bytes;
 - The following byte 'VT', or fourth 'column', contains the variable type (ascii encoding) - (See [Variable Types](#));
 - The next 6 bytes are the "Node From" (ascii encoding);
 - The next 6 bytes are the "Node To" (ascii encoding);
- In 'row' 5 + the number of variables, the actual data is recorded:
 - They are composed of 4 bytes fields for IEEE 754 floating point, little endian;
 - The first 4 bytes are the time for the first simulation step;
 - Then comes the variables data, in the same sequence which the variables are given, for the first time step;
 - The above mentioned sequence repeats for all simulation's steps.
- With the current information, the number of simulations steps can be calculated as: $STEPS = (PL4\ SIZE - 5 \times 16 - \#VAR \times 16) / ((\#VAR + 1) \times 4)$;
 - This information allows to dynamically allocate the array to store the simulations results with the dimensions of STEPS x #VAR+1 (the +1 stands for the time column);

- As consequence, $TMAX = (STEPS-1)*DELTAT$;

▼ Pages 6
Find a page or section...
▼ Home
▼ How to use it.
▼ PISA's PL4 Format Frequency domain
▼ PISA's PL4 Format Time domain Binary file byte's positions for time domain simulations
▼ References
▼ Variables Types

Clone this wiki locally

<https://github.com/ldemattos/readPL4.wiki.git>

